WAKISSHA UACE CHEMISTRY PAPER 2 PROPOSED MARKING / SCORING GUIDE 2026 ITEM 2

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Test	Observation	Deduction
A spatula endful of sample B in a dry test tube was strongly heated until there was no further change using heat source, acc bunsen burner, stove	A pale green powdery solid A colourless gas which turns moist blue litmus paper red and formed white precipitate with calcium hydroxide solution ~~White residue~~ Black residue	Cu^{2+}, Ni^{2+}, Fe^{2+} $CO_{2(g)}$ evolved CO_3^{2-} or $C_2O_4^{2-}$ present $CuO_{(s)}$ formed hence Cu^{2+}; Ni^{2+}
Redish brown residue when hot turned brown on cooling		$PbO_{(s)}$ formed hence Pb^{2+}

Can be ignored and Continue marking

To a spatula endful of sample B in a dry test tube was added 3 drops of Concentrated sulphuric acid and then warmed gently	Rapid effervescence of a colourless gas that turned moist blue litmus paper red and formed a white precipitate with Calcium hydroxide solution.	CO_2 (g) evolved CO_3^{2-} present
	Effervescence of brown vapour which turns moist blue litmus paper red and bleaches it	Br_2(g) evolved Br^- present

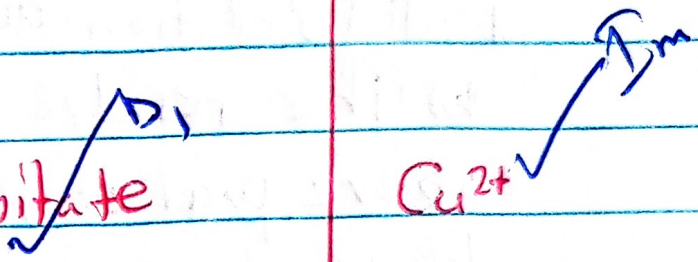
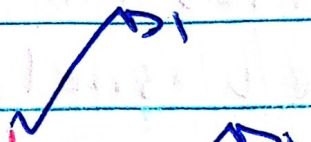
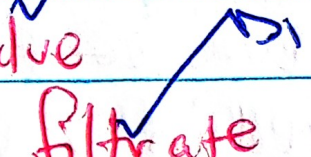
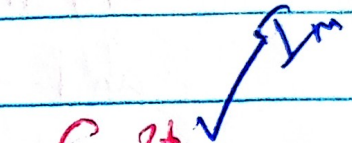
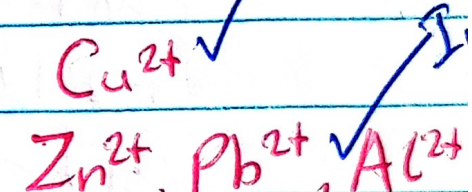
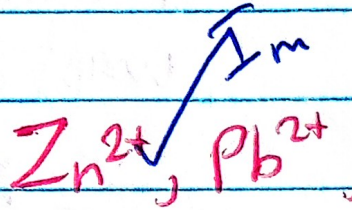
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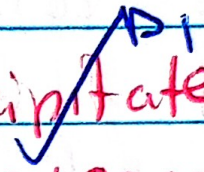
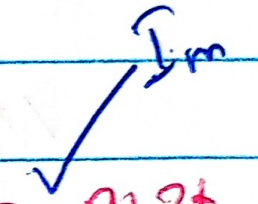
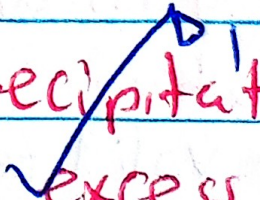
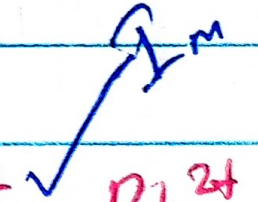
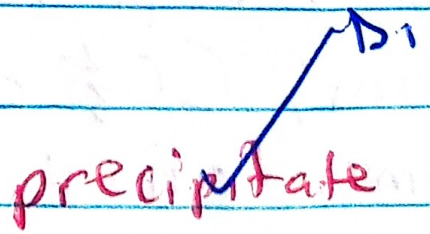
Test	Observation	Deduction
Two spatula endfuls of sample B were dissolved in about $5cm^3$ of distilled water and the mixture shaken throughly.	Partly soluble to form pale green suspension	Ni^{2+}, Cu^{2+}
i. The resultant solution was filtered and both the filtrate and the residue kept	Pale green residue	
	Colourless filtrate	
a) The filtrate was divided into 3 portions		
i) To the first portion was added 3 drops of lead(II) nitrate solution followed by warming.	White precipitate insoluble on warming	Br^-

(ii) To the ~~third~~ second portion was added 3 drops of silver nitrate solution followed by dilute ammonia solution dropwise until in excess.	A cream precipitate, dissolves with difficulty in excess ammonia solution.	Br^- ✓ Im
(iii) To the ~~fourth~~ third part was added 1 drop of dilute nitric acid followed by 1 drop of a bleaching agent solution, and then 3 drops of Chloroform and shaken ~~gently~~ gently. acc other valid tests	An orange solution in the organic layer. ✓ D1	Br^- confirmed present. ✓ Im

D r ≥ 18
D ≥ 2

Im ≥ 25

Test	Observation	Deduction
2. The residue in 1 was dissolved in dilute nitric acid. To the resultant solution, 7cm³ of dilute sodium hydroxide solution was added drop wise and then filtered. Both the filtrate and the residue were kept.	Blue precipitate insoluble in excess	Cu^{2+} 
	Blue residue  Colourless filtrate 	Cu^{2+}  Zn^{2+}, Pb^{2+}, Al^{2+} 
a) To 4cm³ of the filtrate obtained in 1 above was added dilate nitric acid drop by drop untill the solution is just acidic.	White precipitate soluble in the acid to form a Colourless solution.	Zn^{2+}, Pb^{2+}, Al^{3} 

The acidic solution was divided into 4 portions (parts) i) To the first part was added dilute sodium hydroxide solution dropwise until in excess	White precipitate soluble in excess 	Zn^{2+}, Pb^{2+}, Al^{3+} 
ii) To the second part was added dilute ammonia solution dropwise until in excess	White precipitate insoluble in excess 	Al^{3+}, Pb^{2+} 
To the third portion was added dilute sulphuric acid	White precipitate insoluble 	Pb^{2+} 

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Test Procedure	Observation	Deductio
To the fourth part was added 3 drops of potassium Iodid solution acc other valid tests	Yellow precipitate D1	Pb^{2+} 1m confirmed present.
b) The residue in 2 above was washed in dilute ~~ammonia~~ sodium hydroxide solution and the washed residue was ~~cast~~ dissolved in dilute nitric acid. The resultant solution was dissolved into ___ parts.	Effervescence of a colourless gas that turns moist blue litmus paper red and forms white precipitate with Calcium hydroxide solution. D1	$CO_{2(g)}$ 1m evolved CO_3^{2-} Confirmed present
i) To the first part was added dilute sodium hydroxide solution dropwise until in excess	Blue precipitate D1 insoluble in excess	Cu^{2+} 1m

ii) To the second part was added dilute ammonia solution dropwise until in excess	Blue precipitate soluble in excess forming deep blue solution.	Cu^{2+} 3m
iii) To the third part was added 3 drops of potassium iodide solution Acc other valid tests	White precipitate in brown solution	Cu^{2+} 1m confirmed present.

Conclusion: The qualitative and analysis of Sample B confirmed the presence of two cations, ie Pb^{2+} and Cu^{2+}, two anions; CO_3^{2-} and Br^- 1a

Recommendation: The waste should be collected and safely as a harzardous waste, after recovering the valuable copper and lead compounds to prevent environmental contamination 1c 1c